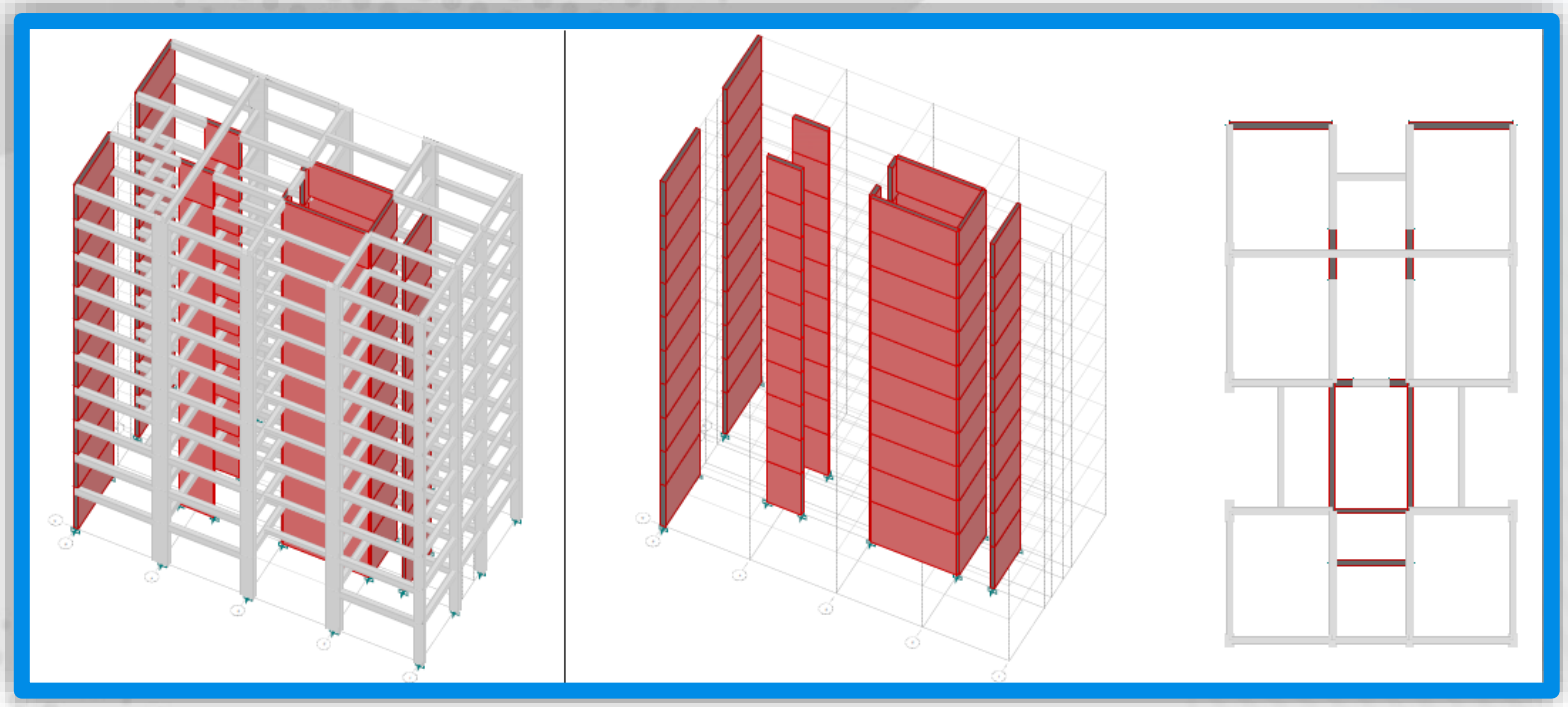


ANÁLISIS Y DISEÑO DE EDIFICACIONES DE CONCRETO ARMADO CON ETABS

SESIÓN 4 – FUERZAS LATERALES EQUIVALENTES

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KONSTRUEDU.COM

PESO SÍSMICO

- REFERENCIAS

NCh433

5.4.2.3 **Sistemas mixtos**

Las cargas gravitacionales y sísmicas son resistidas por una combinación de los sistemas anteriores.

5.5 **Modelos estructurales**

5.5.1 Para el cálculo de las masas se deben considerar las cargas permanentes más un porcentaje de la sobrecarga de uso, que no puede ser inferior a 25% en construcciones destinadas a la habitación privada o al uso público donde no es usual la aglomeración de personas o cosas, ni a un 50% en construcciones en que es usual esa aglomeración.

NCh433

25.7. En los edificios cuyos elementos estructurales predominantes sean muros, se considera un modelo que tome en cuenta la interacción entre muros en direcciones perpendiculares (muros en H, muros en T y muros en L).

Artículo 26.- Estimación del Peso (P)

El peso (P) se calcula adicionando a la carga permanente y total de la edificación un porcentaje de la carga viva o sobrecarga que se determina de la siguiente manera:

a) En edificaciones de las categorías A y B, se toma el 50% de la carga viva.
b) En edificaciones de la categoría C, se toma el 25% de la carga viva.
c) En depósitos, se toma el 80% del peso total que es posible almacenar.
d) En azoteas y techos en general se toma el 25% de la carga viva.

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12.7.2 Effective Seismic Weight. The effective seismic weight, W, of a structure shall include the dead load, as defined in Section 3.1, above the base and other loads above the base as listed below:

1. In areas used for storage, a minimum of 25% of the floor live load shall be included.
   EXCEPTIONS:
   a. Where the inclusion of storage loads adds no more than 5% to the effective seismic weight at that level, it need not be included in the effective seismic weight.
   b. Floor live load in public garages and open parking structures need not be included.
2. Where provision for partitions is required by Section 4.3.2 in the floor load design, the actual partition weight or a minimum weight of 10 psf (0.48 kN/m^2) of floor area, whichever is greater.
3. Total operating weight of permanent equipment.
4. Where the flat roof snow load, P_f, exceeds 30 psf (1.44 kN/m^2), 20% of the uniform design snow load, regardless of actual roof slope.
5. Weight of landscaping and other materials at roof gardens and similar areas.

C12.7.2 Effective Seismic Weight. During an earthquake, the structure accelerates laterally, and these accelerations of the structural mass produce inertial forces. These inertial forces, accumulated over the height of the structure, produce the seismic base shear.

When a building vibrates during an earthquake, only that portion of the mass or weight that is physically tied to the structure needs to be considered as effective. Hence, live loads (e.g., loose furniture, loose equipment, and human occupants) need not be included. However, certain types of live loads, such as storage loads, may develop inertial forces, particularly where they are densely packed.

ASCE 7-16

ANÁLISIS DE FLE

- Análisis Sísmico FLE ASCE 7-16
  1. Determinar la categoría de la edificación (I-IV).
  2. Determinar parámetros básicos de movimiento de terreno (Ss,S1).
  3. Determinar la clasificación de sitio (A-F).
  4. Determinar factores de coeficientes de sitio (Fa, Fv).
  5. Determinar parámetros de sismo de diseño (Sds,Sd1).
  6. Determinar categoría sísmica de diseño (A-F).
  7. Determinar factor de importancia.
  8. Seleccionar sistema estructural y parámetros de sistema (R, Cd, Ωo)
  9. Examinar irregularidades para sistema estructural.
  10. Determinar flexibilidad de diafragma.
  11. Determinar factor de redundancia.
  12. Determinar procedimiento de fuerzas laterales equivalentes.
  13. Cálculo de fuerzas laterales.
  14. Agregar cargas torsionales.
  15. Agregar cargas ortogonales.
  16. Ejecutar análisis.
  17. Combinar resultados.
  18. Revisión de resistencia, deflexión y estabilidad.

ANÁLISIS DE FLE

- 1. Determinar parámetros básicos de movimiento de terreno (Ss,S1).

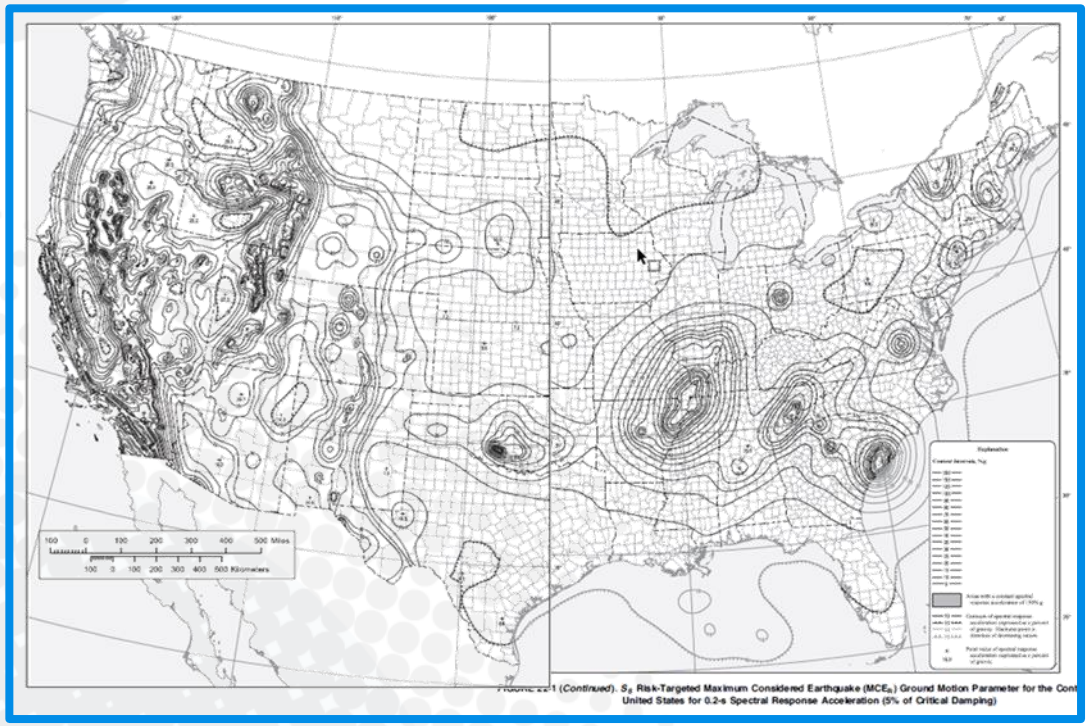
Table 1.5-1 Risk Category of Buildings and Other Structures for Flood, Wind, Snow, Earthquake, and Ice Loads

Use or Occupancy of Buildings and Structures	Risk Category
Buildings and other structures that represent low risk to human life in the event of failure	I
All buildings and other structures except those listed in Risk Categories I, III, and IV	II
Buildings and other structures, the failure of which could pose a substantial risk to human life	III
Buildings and other structures, not included in Risk Category IV, with potential to cause a substantial economic impact and/or mass disruption of day-to-day civilian life in the event of failure	
Buildings and other structures not included in Risk Category IV (including, but not limited to, facilities that manufacture, process, handle, store, use, or dispose of such substances as hazardous fuels, hazardous chemicals, hazardous waste, or explosives) containing toxic or explosive substances where the quantity of the material exceeds a threshold quantity established by the Authority Having Jurisdiction and is sufficient to pose a threat to the public if released[a]	
Buildings and other structures designated as essential facilities	IV

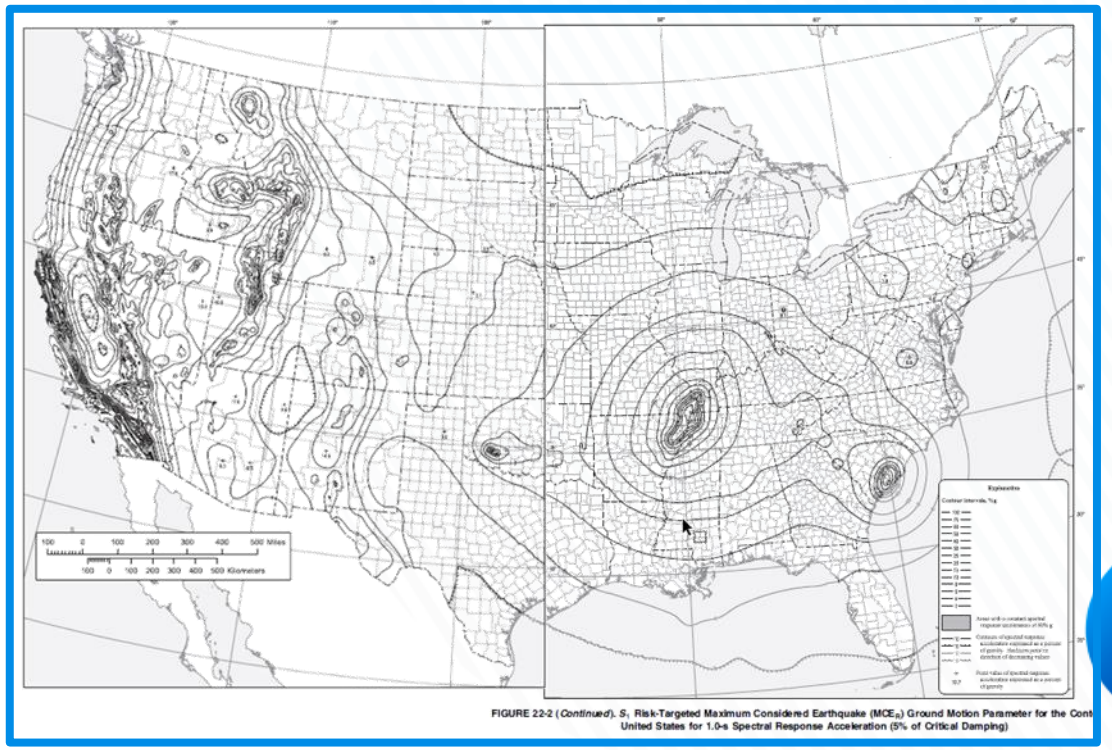
1.5/Table1.5-1 (ASCE 7-16)

ANÁLISIS DE FLE

- Determinar parámetros básicos de movimiento de terreno (Ss,S1).
  - ❑ Obtener Ss y S1 de los mapas “Ground Motion Parameter”
  - ❑ Zona California ZIP code 00012
  - ❑ Ss = 0.934355, S1 = 0.323589



22/Figs 22.1, 22.3, 22.5, 22.6 (ASCE 7-16)



22/Figs 22.1, 22.3, 22.5, 22.6 (ASCE 7-16)

ANÁLISIS DE FLE

- Determinar la clasificación de sitio (A-F).
  - ❑ Suelo de fundación. Roca.
  - ❑ Clase de sitio B

Table 20.3-1 Site Classification

Site Class	$\bar{v}_s$	$\bar{N}$ or $\bar{N}_{ch}$	$\bar{s}_u$
A. Hard rock	>5,000 ft/s	NA	NA
B. Rock	2,500 to 5,000 ft/s	NA	NA
C. Very dense soil and soft rock	1,200 to 2,500 ft/s	>50 blows/ft	>2,000 lb/ft^2
D. Stiff soil	600 to 1,200 ft/s	15 to 50 blows/ft	1,000 to 2,000 lb/ft^2
E. Soft clay soil	<600 ft/s	<15 blows/ft	<1,000 lb/ft^2
	Any profile with more than 10 ft of soil that has the following characteristics: — Plasticity index $PI > 20$, — Moisture content $w \geq 40\%$, — Undrained shear strength $\bar{s}_u < 500$ lb/ft^2		
F. Soils requiring site response analysis in accordance with Section 21.1	See Section 20.3.1		

Note: For SI: 1 ft = 0.3048 m; 1 ft/s = 0.3048 m/s; 1 lb/ft^2 = 0.0479 kN/m^2.

20/Table 20.3-1(ASCE 7-16)

ANÁLISIS DE FLE

- Determinar factores de ajuste coeficientes de sitio (Fa, Fv).

❑ Ss = 0.934355, S1 = 0.323589

$$V = C_s \cdot W \qquad (12.8\text{-}1)$$

$$C_s = 0.044 \cdot S_{DS} \cdot I_e$$

$$C_s = \frac{S_{DS}}{\left(\frac{R}{I_e}\right)} \qquad (12.8\text{-}2)$$

$$C_s = \frac{S_{D1}}{T\left(\frac{R}{I_e}\right)} \qquad T \le T_L \qquad (12.8\text{-}3)$$

$$C_s = \frac{S_{D1} \cdot T_L}{T^2 \cdot \left(\frac{R}{I_e}\right)} \qquad T > T_L \qquad (12.8\text{-}4)$$

$$S_{MS} = F_a \cdot S_s \qquad (11.4\text{-}1)$$

$$S_{M1} = F_v \cdot S_1 \qquad (11.4\text{-}1)$$

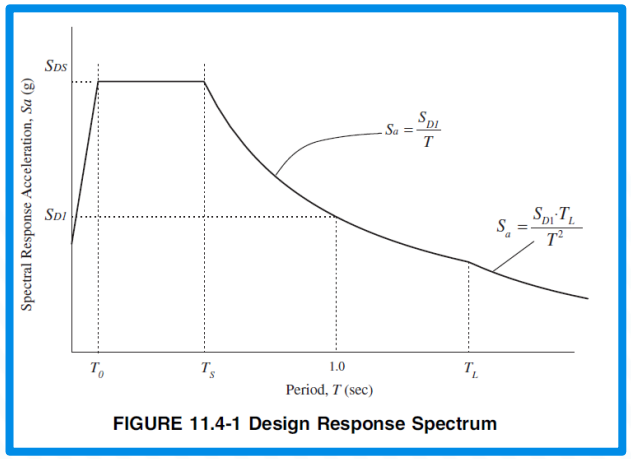


FIGURE 11.4-1 Design Response Spectrum

Table 11.4-1 Short-Period Site Coefficient, F_a

Site Class	Mapped Risk-Targeted Maximum Considered Earthquake (MCE_R) Spectral Response Acceleration Parameter at Short Period					
	$S_S \le 0.25$	$S_S = 0.5$	$S_S = 0.75$	$S_S = 1.0$	$S_S = 1.25$	$S_S \ge 1.5$
A	0.8	0.8	0.8	0.8	0.8	0.8
B	0.9	0.9	0.9	0.9	0.9	0.9
C	1.3	1.3	1.2	1.2	1.2	1.2
D	1.6	1.4	1.2	1.1	1.0	1.0
E	2.4	1.7	1.3	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8
F	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8

Note: Use straight-line interpolation for intermediate values of S_s.

Table 11.4-2 Long-Period Site Coefficient, F_v

Site Class	Mapped Risk-Targeted Maximum Considered Earthquake (MCE_R) Spectral Response Acceleration Parameter at 1-s Period					
	$S_1 \le 0.1$	$S_1 = 0.2$	$S_1 = 0.3$	$S_1 = 0.4$	$S_1 = 0.5$	$S_1 \ge 0.6$
A	0.8	0.8	0.8	0.8	0.8	0.8
B	0.8	0.8	0.8	0.8	0.8	0.8
C	1.5	1.5	1.5	1.5	1.5	1.4
D	2.4	2.2[a]	2.0[a]	1.9[a]	1.8[a]	1.7[a]
E	4.2	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8
F	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8

Note: Use straight-line interpolation for intermediate values of S_1.
[a]Also, see requirements for site-specific ground motions in Section 11.4.8.

20/Table 20.3-1(ASCE 7-16)

ANÁLISIS DE FLE

- Determinar parámetros de sismo de diseño (Sds,Sd1).
  - ❑ Ss = 0.934355, S1 = 0.323589
  - ❑ S_{MS} = 0.8409195, S_{M1} = 0.2588712
  - ❑ S_{DS} = 0.560613, S_{D1} = 0.1725808

$$S_{MS} = F_a \cdot S_s \quad (11.4\text{-}1)$$

$$S_{M1} = F_v \cdot S_1 \quad (11.4\text{-}1)$$

$$S_{DS} = \frac{2}{3} S_{MS} \quad (12.8\text{-}2)$$

$$S_{D1} = \frac{2}{3} S_{M1} \quad (12.8\text{-}2)$$

ANÁLISIS DE FLE

- Determinar categoría sísmica de diseño (A-F).
  - ❑ S_{DS} = 0.560613, S_{D1} = 0.1725808

TABLE 11.6-1 Seismic Design Category Based on Short-Period Response Acceleration Parameter

	Risk Category	
Value of S_{DS}	I or II or III	IV
$S_{DS} < 0.167$	A	A
$0.167 \le S_{DS} < 0.33$	B	C
$0.33 \le S_{DS} < 0.50$	C	D
$0.50 \le S_{DS}$	D	D

TABLE 11.6-2 Seismic Design Category Based on 1-s Period Response Acceleration Parameter

	Risk Category	
Value of S_{D1}	I or II or III	IV
$S_{D1} < 0.067$	A	A
$0.067 \le S_{D1} < 0.133$	B	C
$0.133 \le S_{D1} < 0.20$	C	D
$0.20 \le S_{D1}$	D	D

20/Table 20.3-1(ASCE 7-16)

ANÁLISIS DE FLE

- Determinar factor de importancia.

Table 1.5-2 Importance Factors by Risk Category of Buildings and Other Structures for Snow, Ice, and Earthquake Loads

Risk Category from Table 1.5-1	Snow Importance Factor, I_s	Ice Importance Factor—Thickness, I_i	Ice Importance Factor—Wind, I_w	Seismic Importance Factor, I_e
I	0.80	0.80	1.00	1.00
II	1.00	1.00	1.00	1.00
III	1.10	1.15	1.00	1.25
IV	1.20	1.25	1.00	1.50

Note: The component importance factor, I_p, applicable to earthquake loads, is not included in this table because it depends on the importance of the individual component rather than that of the building as a whole, or its occupancy. Refer to Section 13.1.3.

1.5/Table1.5-1 (ASCE 7-16)

ANÁLISIS DE FLE

- Seleccionar sistema estructural y parámetros de sistema (R, Cd, Ωo)
  - ❑ Ss = 0.934355, S1 = 0.323589
  - ❑ S_{MS} = 0.8409195, S_{M1} = 0.2588712
  - ❑ S_{DS} = 0.560613, S_{D1} = 0.1725808

$$S_{MS} = F_a \cdot S_s \quad (11.4\text{-}1)$$

$$S_{M1} = F_v \cdot S_1 \quad (11.4\text{-}1)$$

$$S_{DS} = \frac{2}{3} S_{MS} \quad (12.8\text{-}2)$$

$$S_{D1} = \frac{2}{3} S_{M1} \quad (12.8\text{-}2)$$

$$C_s = \frac{S_{D1}}{T\left(\frac{R}{I_e}\right)} \qquad T \le T_L \quad (12.8\text{-}3)$$

20/Table 20.3-1(ASCE 7-16)

ANÁLISIS DE FLE

- Seleccionar sistema estructural y parámetros de sistema (R, Cd, Ωo)

Table 12.2-1 Design Coefficients and Factors for Seismic Force-Resisting Systems

Seismic Force-Resisting System	ASCE 7 Section Where Detailing Requirements Are Specified	Response Modification Coefficient, R^a	Overstrength Factor, $\Omega_0{}^b$	Deflection Amplification Factor, $C_d{}^c$	Structural System Limitations Including Structural Height, h_n (ft) Limits[d] — Seismic Design Category B	C	D[e]	E[e]	F[f]
D. DUAL SYSTEMS WITH SPECIAL MOMENT FRAMES CAPABLE OF RESISTING AT LEAST 25% OF PRESCRIBED SEISMIC FORCES	12.2.5.1								
1. Steel eccentrically braced frames	14.1	8	2½	4	NL	NL	NL	NL	NL
2. Steel special concentrically braced frames	14.1	7	2½	5½	NL	NL	NL	NL	NL
3. Special reinforced concrete shear walls[g,h]	14.2	7	2½	5½	NL	NL	NL	NL	NL
4. Ordinary reinforced concrete shear walls[g]	14.2	6	2½	5	NL	NL	NP	NP	NP
5. Steel and concrete composite eccentrically braced frames	14.3	8	2½	4	NL	NL	NL	NL	NL
6. Steel and concrete composite special concentrically braced frames	14.3	6	2½	5	NL	NL	NL	NL	NL
7. Steel and concrete composite plate shear walls	14.3	7½	2½	6	NL	NL	NL	NL	NL
8. Steel and concrete composite special shear walls	14.3	7	2½	6	NL	NL	NL	NL	NL
9. Steel and concrete composite ordinary shear walls	14.3	6	2½	5	NL	NL	NP	NP	NP
10. Special reinforced masonry shear walls	14.4	5½	3	5	NL	NL	NL	NL	NL
11. Intermediate reinforced masonry shear walls	14.4	4	3	3½	NL	NL	NP	NP	NP
12. Steel buckling-restrained braced frames	14.1	8	2½	5	NL	NL	NL	NL	NL
13. Steel special plate shear walls	14.1	8	2½	6½	NL	NL	NL	NL	NL

12/Table 12.2-1-1(ASCE 7-16)

ANÁLISIS DE FLE

- Seleccionar sistema estructural y parámetros de sistema (R, Cd, Ωo)

  - Ss = 0.934355, S1 = 0.323589
  - S_{MS} = 0.8409195, S_{M1} = 0.2588712
  - S_{DS} = 0.560613, S_{D1} = 0.1725808
  - T = 1.238seg, Tl = 8seg
  - W = 785.87tonf
  - Cs = 0.0800
  - Cs_min = 0.0247
  - Cs_max = 0.0199
  - Cs = 0.0247
  - V = 19.41 tonf

$$C_s = 0.044 \cdot S_{DS} \cdot I_e$$

$$C_s = \frac{S_{DS}}{\left(\frac{R}{I_e}\right)} \quad (12.8\text{-}2)$$

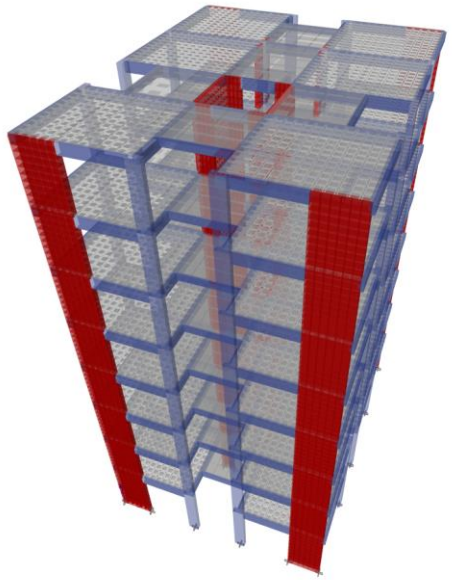
$$C_s = \frac{S_{D1}}{T\left(\frac{R}{I_e}\right)} \qquad T \leq T_L \quad (12.8\text{-}3)$$

12/Table 12.2-1-1(ASCE 7-16)

ANÁLISIS DE FLE (CASO 3D)

- Seleccionar sistema estructural y parámetros de sistema (R, Cd, Ωo)

  - Ss = 0.934355, S1 = 0.323589
  - S_{MS} = 0.8409195, S_{M1} = 0.2588712
  - S_{DS} = 0.560613, S_{D1} = 0.1725808
  - T = 0.8917seg, Tl = 8seg
  - W = 3373.81tonf
  - Cs = 0.0800
  - Cs_min = 0.0247
  - Cs_max = 0.0276
  - Cs = 0.0247
  - V_min = 83.333 tonf
  - V_max= 93.117 tonf



$$C_s = \frac{S_{DS}}{\left(\frac{R}{I_e}\right)} \quad (12.8\text{-}2)$$

$$C_s = 0.044 \cdot S_{DS} \cdot I_e$$

$$C_s = \frac{S_{D1}}{T\left(\frac{R}{I_e}\right)} \qquad T \le T_L \quad (12.8\text{-}3)$$

12/Table 12.2-1-1(ASCE 7-16)

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